only if all other things are equal. We must look to the inequalities if we are to explain why helping at the nest occurs in some species and not others.

Think, for instance, about a species of birds that nest in hollow trees. These trees are precious, for only a limited supply is available. If you are a young adult whose parents are still alive, they are probably in possession of one of the few available hollow trees (they must have possessed one at least until recently; otherwise, you wouldn't exist). So you are probably living in a hollow tree that is a thriving going concern, and the new baby occupants of this productive hatchery are your full brothers and sisters, genetically as close to you as your own offspring would be. If you leave and try to go it alone, your chances of obtaining a hollow tree are low. Even if you succeed, the offspring that you rear will be no closer to you, genetically, than brothers and sisters. A given quantity of effort invested in your parents' hollow tree is better value than the same quantity of effort invested in trying to set up on your own. These conditions, then, might favour sibling care —'helping at the nest'.

In spite of all this, it remains true that some individuals—or all individuals some of the time—must go out and seek new hollow trees, or whatever the equivalent is for their species. To use the 'bearing and caring' terminology of Chapter 7, *somebody* has to do some bearing; otherwise, there would be no young to care for! The point here is not that 'otherwise, the species would go extinct'. Rather, in any population dominated by genes for pure caring, genes for bearing will tend to have an advantage. In social insects the bearer role is filled by the queens and males. They are the ones that go out into the world, looking for new 'hollow trees', and that is why they are winged, even in ants, whose workers are wingless. These reproductive castes are specialized for their whole lifetime. Birds and mammals that help at the nest do it the other way. Each individual spends part of its life (usually its first adult season or two) as a 'worker', helping to rear younger brothers and sisters, while for the remaining part of its life it aspires to be a 'reproductive'.

What about the naked mole rats described in the previous note? They exemplify the going concern or 'hollow tree' principle to perfection, though their going concern does not literally involve a hollow tree. The key to their story is probably the patchy distribution of their food supply underneath the savannah. They feed mainly on underground tubers. These tubers can be very large and very deeply buried. A single tuber of one such species can outweigh 1,000 mole rats and, once found, can last the colony for months or even years. But the problem is finding the tubers, for they are scattered randomly and sporadically throughout the savannah. For mole rats, a food source is difficult to find but well worth it once found. Robert Brett has calculated that a single mole rat, working on its own, would have to search so long to find a single tuber that it would wear its teeth out with digging. A large social colony, with its miles of busily patrolled burrows, is an efficient tuber-mine. Each individual is economically better off as part of a union of fellow miners.

A large burrow system, then, manned by dozens of cooperating workers, is a going concern just like our hypothetical 'hollow tree', only more so! Given that you live in a flourishing communal labyrinth, and given that your mother is still producing full brothers and sisters inside it, the inducement to leave and start a family of your own becomes very low indeed. Even if some of the young produced are only half-siblings, the 'going concern' argument can still be powerful enough to keep young adults at home.

p. 230 They found a rather convincingly close fit to the 3: 1 female to male ratio predicted …

Richard Alexander and Paul Sherman wrote a paper criticizing Trivers and Hare's methods and conclusion. They agreed that female-biased sex ratios are normal among social insects, but disputed the claim that there is a good fit to 3:1. They preferred an alternative explanation for the female-biased sex ratios, an explanation that, like Trivers and Hare's, was first suggested by Hamilton. I find Alexander and Sherman's reasoning quite persuasive, but confess to a gut feeling that a piece of work as beautiful as Trivers and Hare's cannot be all wrong.